

Design of a Fully Differential Two-Stage CMOS Op-Amp for High Gain, High Bandwidth Applications

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Abstract

In this paper design and implementation of a two stage fully differential, RC Miller compensated CMOS operational amplifier is presented. High gain enables this circuit to operate efficiently in a closed loop feedback system, whereas high bandwidth makes it suitable for high speed applications. The design is also able to address any fluctuation in supply or dc input voltages and stabilizes the operation by nullifying the effects due to perturbations. Implementation has been done in 0.18 μm technology using libraries from tsmc with the help of tools from Mentor Graphics and Cadence. Op-amp designed here exhibits >95 dB DC differential gain, ~ 135 MHz unity gain bandwidth, phase margin of $\sim 53^\circ$, and ~ 132 V/uS slew rate for typical 1 pF differential capacitive load.

The power dissipation for 3.3V supply voltage at 27°C temperature under other nominal conditions is 2.29mW. Excellent output differential swing of 5.9V and good liner range of operation are some of the additional features of design.

1. Introduction

The implementation of high performance signal processing and signal conditioning block is one of the most important task in real-time system designing. Operational amplifiers (Op-amps) are one such among various essential components of any kind of signal processing task ranging from simple amplification of weak signals to complex audio and video processing applications in mixed-signal domain. In past few decades CMOS implementation of these building blocks proved superior to its counterparts. The critical issues in this implementation lie in process variations and mismatching of devices and components [1]. Some special layout techniques help us to overcome on the issues related with mismatches during layout of CMOS circuits.

The designing of op-amps puts new challenges in low power applications with reduced channel length devices. Advancements which have appeared recently through new techniques and technologies, give us multiple alternatives in implementations. Involvement of Design Automation (DA) tools in analog and mixed signal design is still not matured as it is in the digital design domain. Accommodation of short channel effects in DA for mixed-signal design is also challenging task for EDA designers.

Here for high gain and high bandwidth applications the well known and enough matured fully differential topology has been employed. For biasing purpose we have designed a current mirror circuit which utilizes a single current source and makes this design free from biasing voltage sources. Post layout simulation result shows that the DC differential gain of 95.278 dB, 135.34 MHz unity gain frequency, 52.8° phase margin, and 131.74 V/uS slew rate are some of the quantitative figure of op-amp designed here. Section 2 briefs about art of op-amp design whereas section 3 explains the analysis of each building block of a two stage op-amp. Design principles behind the design are in section 4. Section 5 talks about the circuit analysis and implementation. Simulation results are presented in section 6 and finally section 7 concludes the work with future directions and improvements.

2. Art of Op-Amp design

High gain in op-amps is not the only desired figure of merit for all kind of signal processing applications. Simultaneously optimizing all parameters has become mandatory now a day in op-amp design. In past few years various new topologies have evolved and have been employed in various applications. Most of them have been also integrated with the existing ones, thus the combination of the two or more resolved the problems which had been noticed earlier in the designs.

Also complete design of modern Op-amps is not only about designing amplification blocks. It also includes the suitable and efficient biasing circuits for all the transistors. Other than this the proper compensation techniques should be employed whenever there is a fear of unstable operation. In two-stage CMOS Op-amps because of two dominant poles the phase margin could easily reach to less than the amount which is just enough for stable operation. This serious problem should be taken care of by designers, otherwise there is a good possibility that the Op-amp output will oscillate and instead of an amplifier it will become an oscillator. In some applications the gain and/or the output swings provided by cascode op-amps are not adequate. In such cases, we resort to "two-stage" op-amps, with the first stage providing a high gain and the second, large swing [2]. In contrast to cascode op-amps, a two-stage configuration isolates the gain and swing requirements as well.

3. Block diagram of two-stage Op-Amp

Generic block diagram with all the basic building blocks of simple two-stage Op-amp is shown below. Each box in the figure can be replaced with an actual circuit implemented in modern VLSI technology. The topologies which a designer chooses are very much dependent upon the type of application he or she is looking for. However it may also depend on some particular advantages and disadvantages of a particular topology.

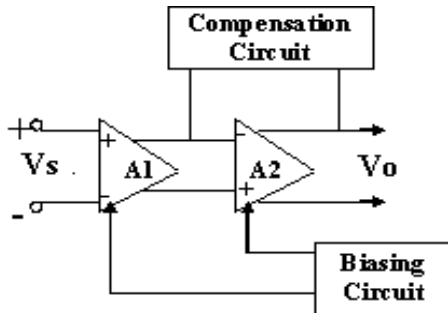


Fig. 1. Generic block diagram of two-stage Op- amp

Here we have chosen simple differential pair amplifier (high noise immune) for input amplifier A1, common source amplifier (high gain) for output amplifier A2, a current mirror circuit (free from voltage sources; utilizing single current reference source) as a biasing circuit, and RC Miller frequency compensation circuit (R and C are in series across the output amplifier). All the basic building blocks are explained below under the sub sections.

3.1. Amplifiers (A1 and A2)

Amplification is an essential function in most analog (and many digital) circuit. Here in this design for input amplifier (A1) a fully differential (in and out) pair with current mirror biasing has been employed. An important advantage of differential operation over single-ended is higher immunity to "environmental" noise. For output stage a common source amplifiers has been used, which is able to provide a large gain in output stage. The advantage of the simple common source (CS) amplifier over differential pair (Diff – amp) is high output swing [3].

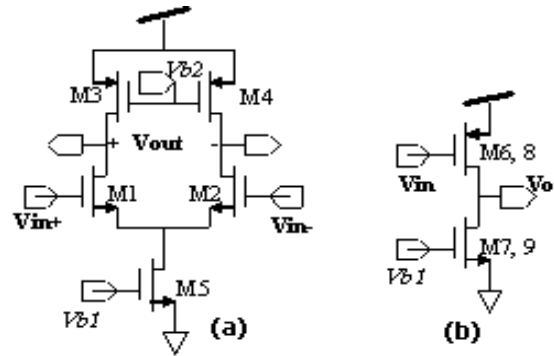


Fig. 2. Input and output stage amplifiers;
(a) Differential pair amplifier (A1), and
(b) Common Source Amplifier (A2)

In A1 stage, M1 and M2 are input NMOS devices whose transconductance appears in gain expression. We keep these devices enough wide operating with small overdrive voltage so that they can produce high gain and high Input Common Mode Rejection ratio (ICMR). The PMOS devices M3 and M4 should exhibit high output resistance when we want high gain, thus we keep them enough long. Tail transistor M5 should have roughly twice overdrive voltage as compare to input devices. This ensures the dimensions of all transistors are roughly of same order. The output intrinsic resistances of M1 or M2 should be roughly equal to the M3 or M4 to get high effective output resistance, thus the high DC gain.

In common source output amplifier (A2) pair the transconductance of input PMOS device M6 or M8 should have a larger value and the NMOS device M7 or M9 should have larger value of output resistance. Equalizing of output resistance of devices will optimize the overall effective resistance, thus the gain of overall system. Biasing voltages Vb1 and Vb2 will be derived by the current mirror circuit by deciding the dimensions of devices used in biasing circuit, which is explained in next sub section.

3.2. Biasing circuit

There are various techniques available in Integrated Circuit (IC) technology for biasing the transistors. Few decades ago voltage biasing was primarily in use, but now a day it is the current biasing which is dominating, due to its various advantages. MOS devices when operate in saturation region, their current is almost constant (neglect lambda effect). A voltage generally produces flow of electron in a material (metal and semiconductor). Same way when a current flows through a material it produces voltage across it. This concept is the core of current mirror circuits.

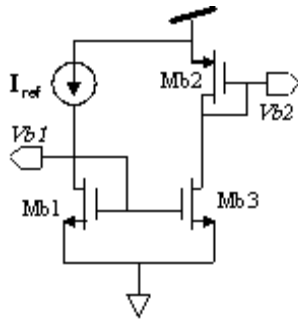


Fig. 3. Current mirror circuit for biasing purpose

In this current mirror circuit the I_{ref} is a current source which is not designed here and assumed to be an idle current source. The Mb1 and Mb3 are NMOS devices whose aspect ratios will be decided based on the requirement of bias voltage $Vb1$ for NMOS transistors (M5, M7 and M9). The PMOS mirror device Mb2 will generate the bias voltage $Vb2$ for PMOS devices (M3 and M4). Mismatch in devices and process variation can easily result into poor performance of circuit and degradation of one or more design parameters [4].

3.3. Frequency compensation circuit

Negative feedback has got so many applications in analog and mix-signal domain. Feedback systems, however, suffer from potential instability, that is they may oscillate. Whenever an amplification stage is introduced in amplifiers it also introduces an additional pole in closed loop system. Due to this additional pole the phase falls drastically if the location of the new pole is of order of less than 10 times of the dominant pole present in system. After phase crossover (PX) point any small amount of feedback will be added to the input, thus will saturate the output. Even if the phase margin is not negative the problem of oscillation or ringing may degrade the performance. So for a

stable operation we need a good amount of phase margin. In most of the cases 60 degree phase margin is considered an optimum one.

There are various techniques of frequency compensation which generally remove or nullify the effects of the poles in frequency response. Pole splitting miller compensation, self compensating capacitor, feed forward compensation using an additional amplifier, negative miller compensation [5] are some of the techniques of frequency compensation. We have used a RC miller compensation technique which is described in following paragraphs.

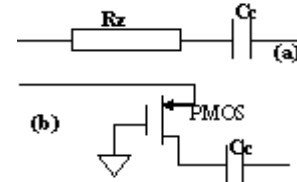


Fig. 4. Miller compensation: (a) A simple series RC implementation. (b) Realization of R_z using a PMOS device whose gate is permanently connected to ground.

Fabrication of large resistances is not preferable in modern VLSI CMOS technology, because they consume a lot of area in silicon. By the characteristics of MOS devices we know that when a MOS device operates in triode region, behaves like a resistor. In triode region the current through and voltage across the MOS are linearly proportional to each other, thus this region of operation is also known as liner region. To take the advantage of this behavior, a PMOS device is being used in this design which consumes very less area in silicon as compared to a resistor. The gate of PMOS has been connected to the ground to ensure that this device will always work in triode region because the source-gate voltage is always enough higher than source-drain voltage in magnitude to keep it in liner region.

The op – amp design takes all the parameters into account which contribute in performance of overall system. High gain, low output impedance, high bandwidth, high output swing, good Power Supply Rejection Ratio (PSRR) and good Common Mode Rejection Ratio (CMRR) are some of the desired features of a good op-amp [6]. Also a design independent from voltage biasing sources is more advantageous than the one with voltage sources. Designers may avoid the optimization of some parameters in initial phase and optimize those only which are going to play major role in performance but at some point of time one has to take care of most of the parameters for a good and versatile design.

4. Design principles and theory

The DC gain expression of a multistage Op- Amp i.e. a two-stage op-amp is

$$A_v = A_{v1} * A_{v2} \quad (1)$$

where A_{v1} or A_{v2} are gain of two different stages for a two stage op-amp can be represented in following form

$$A_{vi} = G_{mi} * R_{outi} \quad (2)$$

here A_{vi} is DC gain of i^{th} stage of Amplifier system independent from the frequency, G_{mi} is transconductance of input network and R_{outi} is the effective output resistance of output network. The complete circuit analysis tells us that

$$G_{m1} = g_{m1,2} \quad (3)$$

$$G_{m2} = g_{m6,8} \quad (4)$$

here the $g_{m1,2}$ is the transconductance of NMOS input transistor M1 or M2 and $g_{m6,8}$ is transconductance of PMOS transistor M6 or M8.

$$R_{out1} = r_{o2} \parallel r_{o4} \quad (5)$$

$$R_{out2} = r_{o6,8} \parallel r_{o7,9} \quad (6)$$

here the r_{o2} and r_{o4} are the output resistances M2 and M4, whereas the $r_{o6,8}$ and $r_{o7,9}$ are the output resistances M6 or M8 and M7 or M9 transistors. Further analysis tells that g_m is a function of device dimension (W/L), bias current (I_D) and overdrive voltage (V_{ov}).

$$g_m = f(W/L, I_D, V_{ov}) \quad (7)$$

It is also a function of process parameters e.g. oxide capacitance C_{ox} and mobility of electrons μ_n . The designers generally do not have control over C_{ox} and μ_n , thus (1) to (7) serve as guiding principles.

5. Circuit analysis and implementations

The complete circuit of the design is as following.

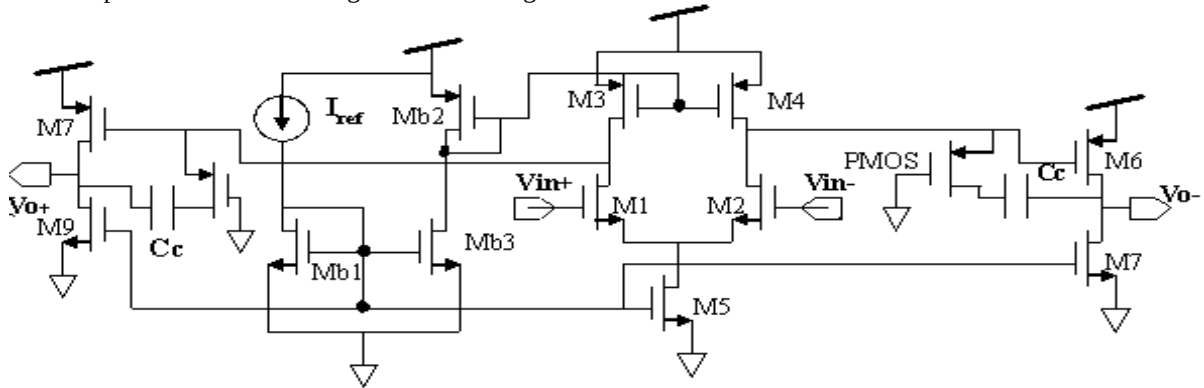


Fig. 5. Schematic of op-amp with compensation

5.1. Power budget

For a supply voltage of 3.3V we started with an initial power budget of <2.5mW, which gives total biasing current of ~750uA. *This is the total current from rail to rail which should be divided into five branches.* After doing a careful analysis for good SR and good UGB we decided to distribute 300uA for differential input amplifier pair, 150uA for single side common source output stage amplifiers and 50uA-50uA for each branch of current mirror circuit.

5.2. Output swing

Here we are targeting initially 5V output differential swing, whereas total output swing is equal to twice of ($V_{dd} - V_{od6} - V_{od7}$). Therefore, $V_{od6} + V_{od7} \leq 0.8V$, because. *Generally, the overdrive of PMOS should be higher than NMOS as mobility of PMOS is approx. 2.5 times less than NMOS.* We decided to start with $V_{od6} = 0.3V$, $V_{od7} = 0.2V$ after a careful analysis.

5.3. Aspect ratios (W/L)

Initial W/L values (in um) can be chosen by using the current expression in saturation region operation. We assumed $\mu_n * C_{ox} = 150 \text{ uA/V}^2$ and $\mu_p * C_{ox} = 60 \text{ uA/V}^2$ for first iteration. The saturation region current expression (8) helped us in calculating the aspect ratios (W/L) of transistors as the current through them is known and overdrive voltage is assigned.

$$I_D = \frac{1}{2} \mu_{n,p} C_{ox} \left(\frac{W}{L} \right)_{n,p} (|V_{gs}| - V_T)^2 \quad (8)$$

here I_D is biasing current, μ_n and C_{ox} are process parameters, W/L is aspect ratio of a transistor, V_{gs} is gate-source voltage and V_T is threshold voltage of device.

The following two rules of thumb we kept in mind while choosing the W/L of MOS devices. First, *Lambda of PMOS is half of the NMOS so the L of PMOS should be roughly double of NMOS to get the equal R_p and R_n for maximum possible gain.* Second, *for the high gain the G_m of input transistors should be high, thus the W/L should be also high.* After taking the above written fact into account and the high R_{out} of the transistors for high effective gain we did new sizing of transistors which was based on many careful iterations.

5.4. Phase margin

Due to High gain the phase margin without compensation was only 3 degree. To improve this we employed an RC compensation technique. We iterated the values of R_z and C_c which gave around 53 degree phase margin. The initial R_z value can be taken from (9).

$$R_z = g_{m6}^{-1} \quad (9)$$

Here some of the noticeable facts are: *Increment in C_c value improves the PM, whereas Increment in R_z value improves the UGB.* To realize the R_z of 2.5 Kohm a PMOS transistor operating in linear region of size (7.3/ 0.72) $\mu\text{m}/\mu\text{m}$ had been used. The area consumed by PMOS to realize a resistance is approximately 10 times lesser than the area consumed by actual implementation of resistance of same value.

6. Simulation results and layout of design

Table 1 below shows the results which had been obtained after post layout simulations. The discrepancies in circuit simulation and post layout simulation are less than 0.5%.

Table 1. Design parameters after layout simulation

Design Specifications	Value (Targeted)	Value (Obtained)
Technology	tsmc018	Used
Supply (V_{dd})	3.3V	3.3V
Load Caps (C_L)	1pF (diff.)	1pF (diff.)
Power Dissipation	$\leq 2.5\text{mW}$	2.288 mW
DC gain (A_v)	$\geq 95\text{ dB}$	95.278 dB
BW (UGB)	$\geq 130\text{ MHz}$	135.34 MHz
Phase Margin	$\geq 55\text{ degree}$	52.8 degree
Output Swing	$\geq 5\text{ V (diff.)}$	5.9 V(diff.)
Slew rate (SR)	$\geq 100\text{ V}/\mu\text{s}$	131.74 V/ μs
CMRR	$\geq 125\text{dB}$	$\sim\text{Inf.}$
PSRR	$\geq 125\text{ dB}$	$\sim\text{Inf.}$
ICMR	$\geq 1.5\text{V}$	1.4 V
Linear range	$\geq 1.5\text{ V}$	1.55 V

The figure 6 shows the frequency response of op-amp after frequency compensation. The plot shows the single side output gain which is 89.46 dB.

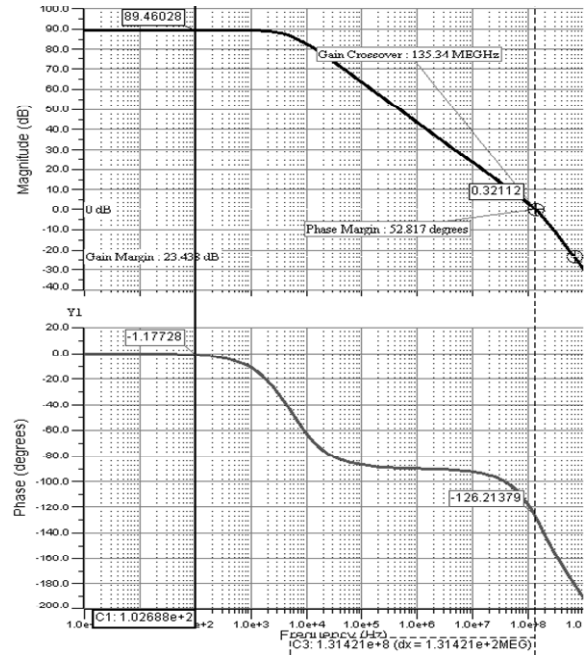


Fig. 6. Magnitude and Phase plot of two stage op-amp

The plot below is the FFT of input and output voltage signals drawn versus frequency. Here the shape of input and output's frequency spectrum are exactly the same in shape. This implies that there is no signal distortion. Also the signal component at 1 KHz is more than 50dB from noise components.

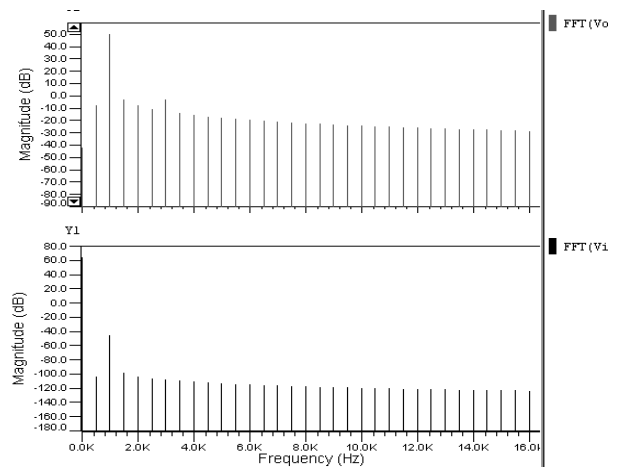


Fig. 7. FFT plot of 1 KHz input and output signals

We made the layout using virtuoso layout editor from Cadence which is shown in fig. 8. Mismatches had been taken care by fingered layout of matched devices [7].

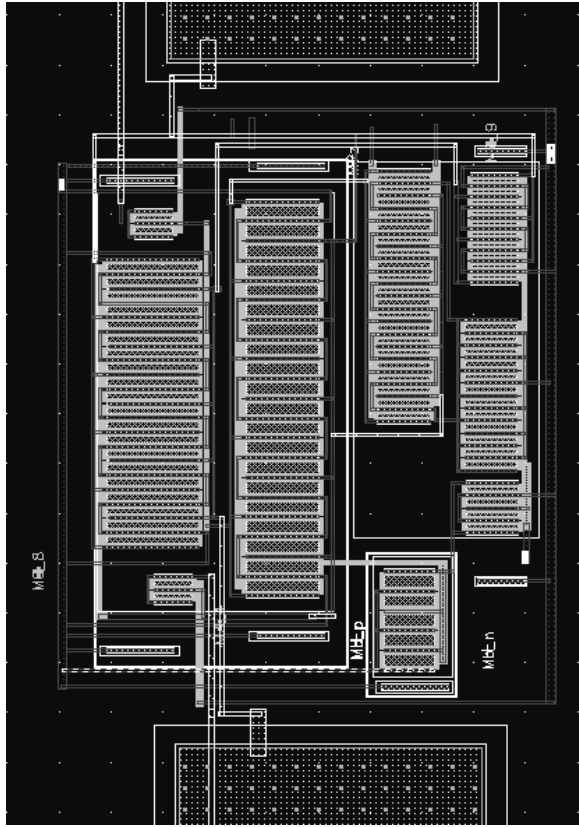


Fig. 8. Layout of two stage op-amp. Only a part of compensation capacitors is visible.

7. Conclusion

Designing of two-stage op-amps is a multi-dimensional-optimization problem where optimization of one or more parameters may easily result into degradation of others. Also the gain bandwidth product which is constant puts challenges to the designers in designing the circuits for high DC gain and high bandwidth applications. Here the gain has been increased by employing thin and long transistors into the design at output stage and wide transistors in input stage. These two techniques are able to increase the gain up to a great extent by increasing the output resistance and input trans-conductance respectively.

Here the improvement in unity gain bandwidth has been done by increasing the bias current which decreases the DC gain and increases power dissipation little bit, still provides a good alternative control to increase bandwidth. Introduction of each stage in multi-stage op-amps exhibits an additional pole into the

system which can create problems in stability. Thus a proper compensation technique has to be employed in the system internally or externally. For this reason the RC-Miller compensation technique has been employed. Fabrication of huge resistor in modern VLSI technology could be another problem which needs to be taken care of. This particular problem has been solved by realizing the series resistor for compensation using a PMOS always operating in triode region [8].

This design does not use any kind of external voltage source for biasing, thus reducing the packaging cost by reducing additional pins for DC bias voltage sources. A simple looking op-amp design problem becomes a harder one when it comes to optimizing all the parameters at a time. A careful analysis of circuit and deep insight into the circuit topologies and device operations leads to good implementation and desired results.

8. Acknowledgment

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9. References

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